IS107 | Power BI Desktop

Laboratory Work 6

# Detecting Anomalies in Time-Series Data

Using Power BI Desktop

Submitted By: Edmundo C. Ederio Jr

Tourism Dataset · Monthly Analysis · Anomaly Detection

# Overview

Dataset

## LW6-tourism-details.csv

920 rows · Monthly tourism data (2012–2022)

Fields

## 4 Columns

Period (Date) · Accommodation Type (Text) · Travel (Text) · Tourists (Number)

Tool

## Power BI Desktop

Line Chart + Analytics Pane Anomaly Detection

Objective

## Detect & Explain Anomalies

Identify and explain outliers in tourist volume over time using AI-powered insights

PART 1

# Data Preparation

*Import LW6-tourism-details.csv and set correct field types in Power BI Desktop.*

**1**

### Get Data → Text/CSV

Click Get Data in the Home ribbon, select Text/CSV, and browse to the tourism CSV file.

**2**

### Period → Date

In Column Tools, change the data type to Date so Power BI recognizes it as a time field.

**3**

### Accommodation Type → Text

Confirm the Accommodation type column is set to Text data type.

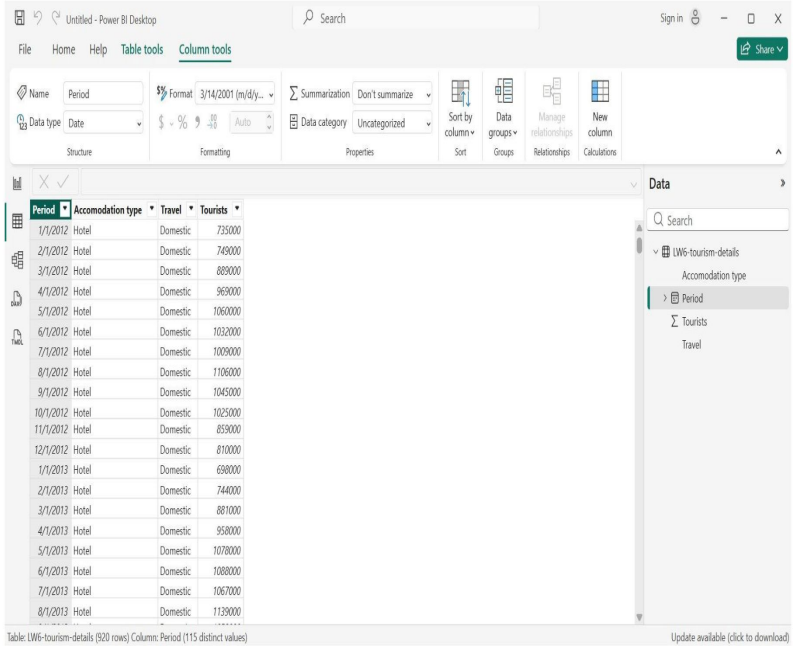
**4**

### Tourists → Whole Number

Set the Tourists column to Whole Number to enable numeric aggregation.

Field: Period → Date | Accommodation Type → Text | Travel → Text | Tourists → Whole Number

Imported LW6-tourism-details.csv into Power BI. The Period column was set to Date type.



PART 2

# Line Chart: Sum of Tourists by Month

*Create a line chart with Period on the X-axis and Tourists on the Y-axis at monthly level.*

### 4 Insert Line Chart

Go to the Visualizations pane and click the Line Chart icon to add the visual.

### 5 Period → X-Axis

Drag the Period field to the Axis well. Expand the date hierarchy.

### 6 Tourists → Values

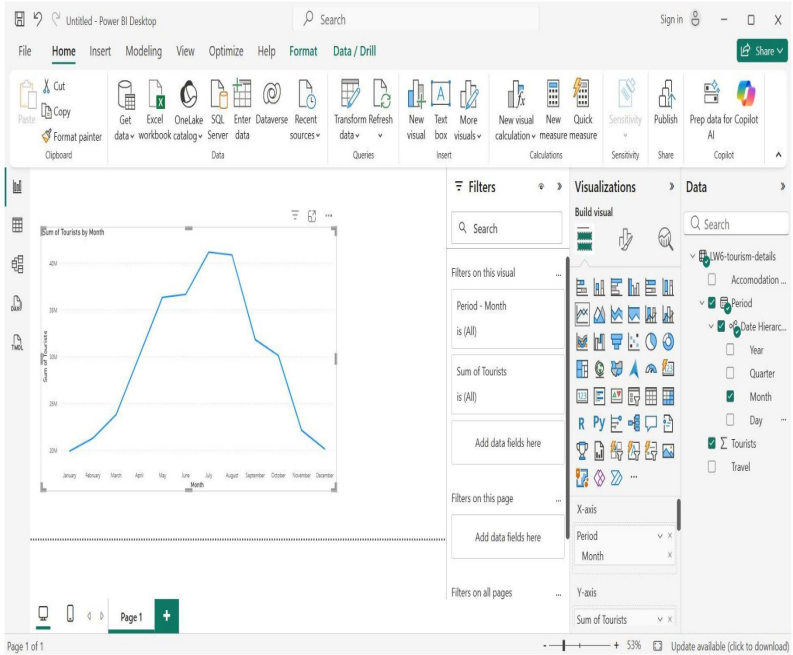
Drag the Tourists field to the Values well to plot tourist count.

### 7 Set Month Level

In the date hierarchy options, select Month to display monthly granularity.

💡 The chart will show seasonal peaks mid-year and dips in winter months across 2012–2022.

A line chart was created with Period on the X-axis and Tourists on the Y-axis (monthly view).



PART 3

# Anomaly Detection Enabled

*Enable anomaly detection in the Analytics pane and set sensitivity to 70%.*

**8**

### Open Analytics Pane

Click the Analytics icon (magnifying glass) in the Visualizations panel on the right side.

**9**

### Toggle Anomalies ON

Scroll to the Anomalies section. Toggle the switch to ON and click Add.

**10**

### Set Sensitivity: 70%

Drag the Sensitivity slider to 70%. This balances detection accuracy.

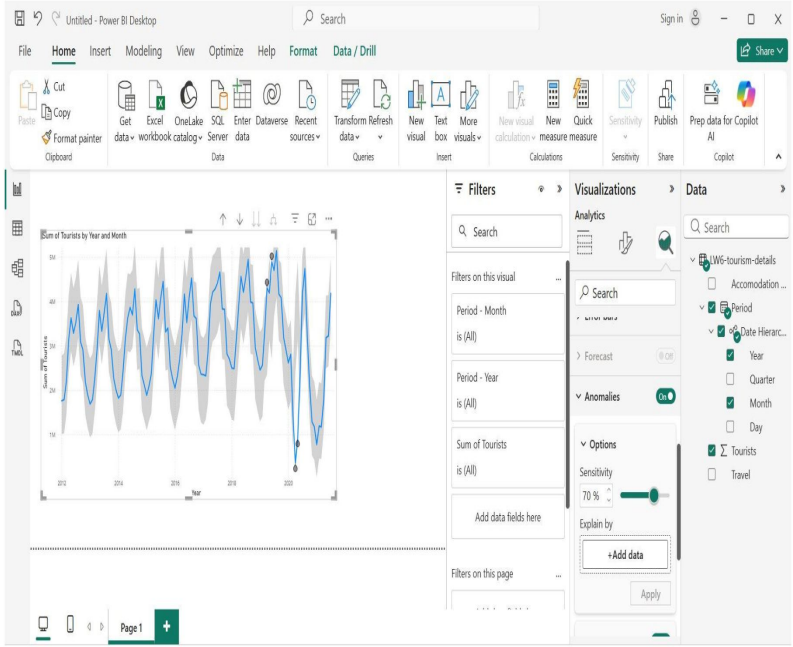
**11**

### Observe the Chart

Gray shaded bands appear showing the expected value range. Anomaly dots are marked.

Sensitivity 70% = balanced detection. Lower = more anomalies. Higher = fewer, stronger ones.

Anomalies toggle turned ON in the Analytics pane. Sensitivity set to 70%. Shaded bands show expected range.



PART 3

# Anomaly Points Highlighted (2019–2021)

*Anomaly markers appear around 2019 (unexpectedly high) and 2020–2021 (sharp drops — COVID-19 impact).*

## 2019 Anomaly

**HIGH ↑**

Tourist numbers surged unexpectedly above the upper bound of the shaded expected range — driven by unusually high Holiday Park bookings.

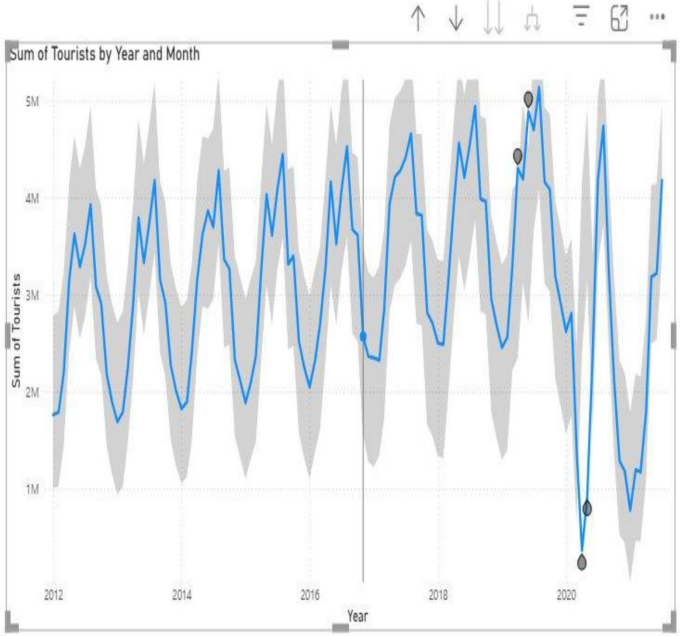
## 2020–2021 Anomaly

**LOW ↓**

Sharp dramatic drops far below the expected band — directly caused by COVID-19 pandemic restrictions on international travel and hotel stays.

📌 The 2020 crash is obvious. The 2019 peak is subtle — just above the upper confidence band.

Anomaly markers appear around 2019 (unexpectedly high) and 2020–2021 (sharp drops — COVID-19 impact).



PART 4

# Explain By: Accommodation Type & Travel

*Drag variables into the Explain By section so Power BI can suggest possible anomaly causes.*

## Step 12 — Add Explanatory Variables

### Accommodation Type

*Hotel · Holiday Park · Camping*

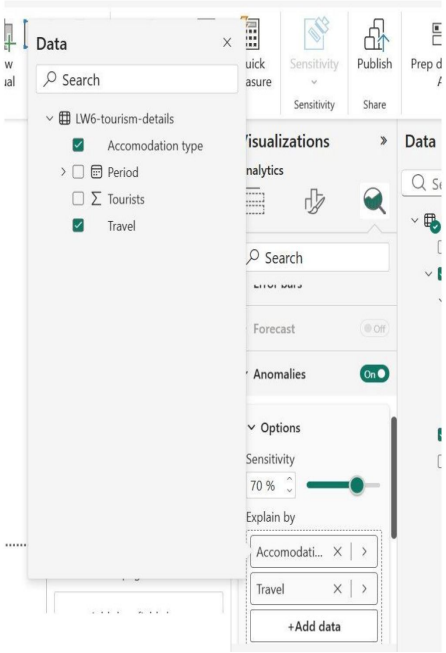
### Travel

*Domestic · International*

## Step 13 — Click an Anomaly to Analyze

- Check direction — Is the value unexpectedly HIGH or LOW?
- Read strength % — Higher % = stronger link to the anomaly.
- Note the driver — Which variable is flagged as unusually high or low?

'Accommodation type' and 'Travel' were dragged into the Explain By section to provide possible explanations.



ANOMALY DETAIL

# Anomaly: Unexpectedly HIGH — June

*4,895,000 tourists exceeded the expected range of 2,707,847 – 4,526,153.*

| 4,895,000 | 4,526,153 | 2,707,847 |
|---|---|---|
| Actual Value | Upper Bound | Lower Bound |

## Possible Explanations

**Holiday Park** HIGH

Sum of tourists for Holiday Park accommodation was unusually high, which may have lifted the overall total.

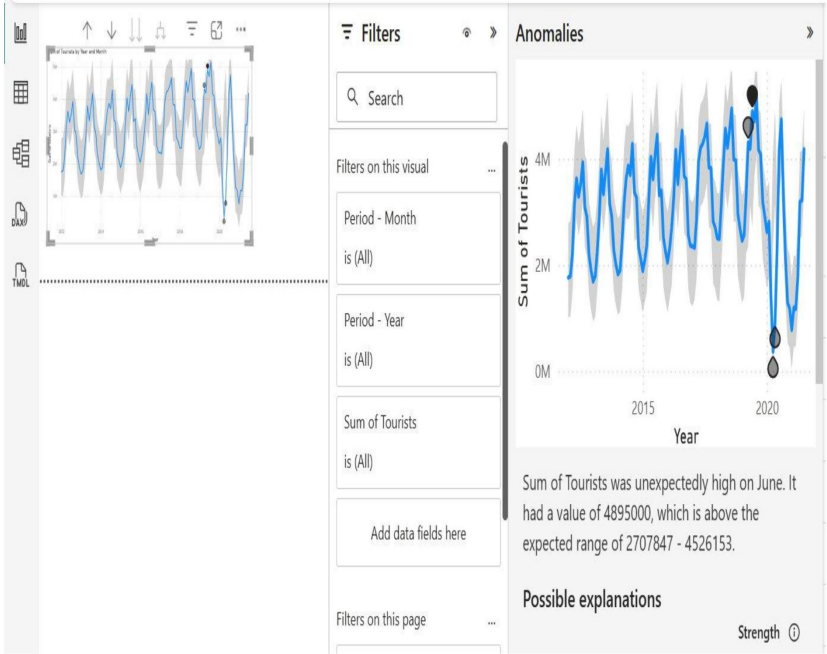
**Hotel Stays** LOW

Hotel accommodation was unusually low — partially offsetting the Holiday Park surge.

**International** LOW

International travel was unusually low, indicating the surge was primarily domestic in origin.

Value of 4,895,000 exceeded the expected range of 2,707,847 – 4,526,153. Triggered the anomaly flag.



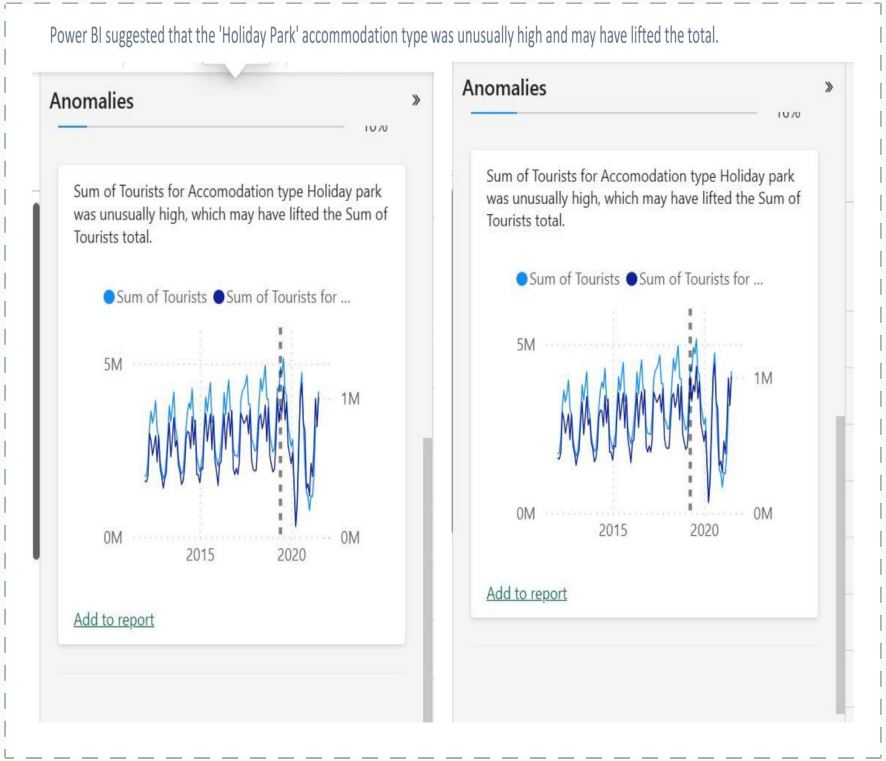
EXPLAIN BY

# Explanation: Holiday Park — Unusually High

*Power BI identified Holiday Park accommodation as unusually high, possibly lifting the total tourist count.*

## What Power BI Found

1. The 'Sum of Tourists for Accommodation type Holiday Park' was flagged as unusually high.
2. This spike in Holiday Park visitors coincided with the June anomaly period.
3. The sub-chart within the Anomalies pane showed Holiday Park trend spiking at the anomaly point.
4. This suggests domestic travelers switched from hotels to holiday park stays during that period.



EXPLAIN BY

# Explanations: Hotel (Low) & International Travel (Low)

*Both Hotel accommodation and International travel were identified as unusually low contributors.*

## Hotel — Unusually LOW

**Strength: 30%**

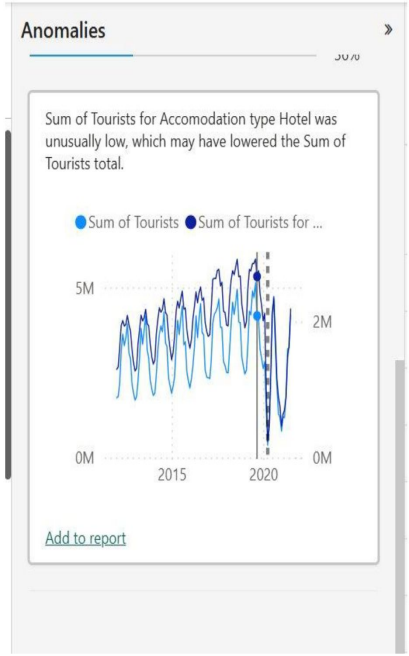
Hotel accommodation was significantly below normal during the anomaly period. Tourists appear to have shifted to Holiday Park stays instead of hotels.

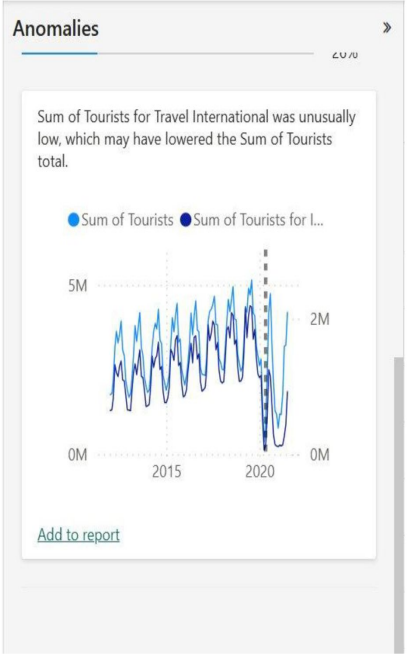
## International Travel — Unusually LOW

**Strength: 20%**

International tourist arrivals were unusually low, suggesting the surge in overall numbers was driven entirely by domestic visitors, not foreign tourists.

Both 'Hotel' accommodation and 'International' travel were identified as unusually low contributors.



PART 5

# Interpretation and Analysis

*Answer the following guide questions based on your Power BI report.*

**Q1** **What trend and seasonality patterns are observed?**

Tourist numbers show consistent peaks mid-year (June–August) and dips in winter. An overall upward trend from 2012–2019 is visible.

**Q2** **Which anomalies are obvious vs. subtle?**

The 2020 COVID-19 crash is highly obvious — a dramatic drop far outside the expected band. The June 2019 peak is subtle — it only slightly exceeds the upper bound.

**Q3** **What factors contributed to anomalies?**

The 2019 high was linked to unusually high Holiday Park bookings. The 2020–2022 lows were driven by COVID-19 — drops in International travel and Hotel stays.

**Q4** **Are the AI explanations logically valid?**

Yes. The suggestions align with real-world context. COVID-19 restrictions directly impacted hotel stays and international travel, confirming the AI's explanations.

PART 6

# After Adding the New Anomaly Dataset

*New 20-row dataset (2021–2022) appended. Report refreshed — new anomaly marker detected in early 2022.*

### 14 Append New Dataset

Add the 20-row anomaly dataset (2021–2022) to your existing CSV. Re-import or use Append in Power Query.

### 15 Refresh the Report

Click Refresh in the Home ribbon. Power BI re-runs anomaly detection on the updated data.

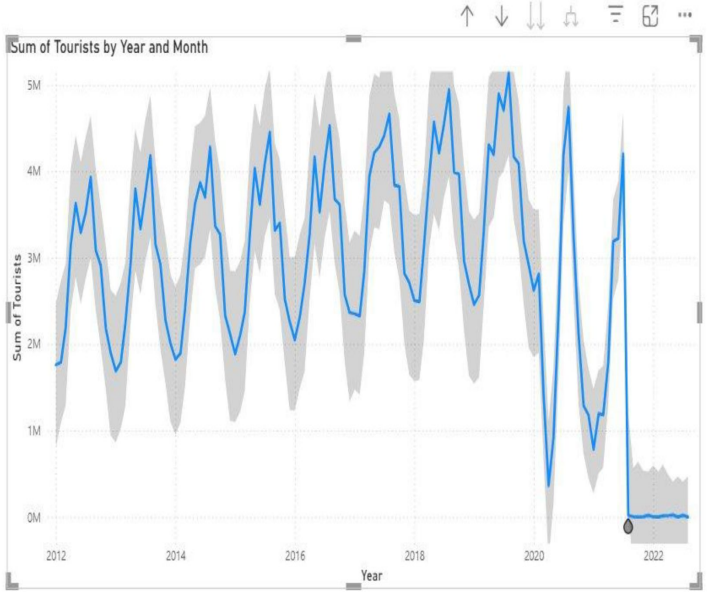
### 16 Observe New Anomalies

A new anomaly marker appears in early 2022. Compare the chart before and after the refresh.

📌 **Key Result:**

Aug 2022 — 20,000 tourists fell far below expected range (61,870 – 1,206,871). Flagged as LOW anomaly.

New 20-row dataset (2021–2022) appended. Report refreshed — new anomaly marker detected in early 2022.



PART 6

# New Anomaly — August 2022: Unexpectedly LOW

*Value of 20,000 tourists fell far below the expected range of 61,870 – 1,206,871.*

| 20,000 | 61,870 | 1,206,871 |
|---|---|---|
| Actual Value | Lower Bound | Upper Bound |

## Possible Explanations

### Domestic Travel

**Strength: 95%**

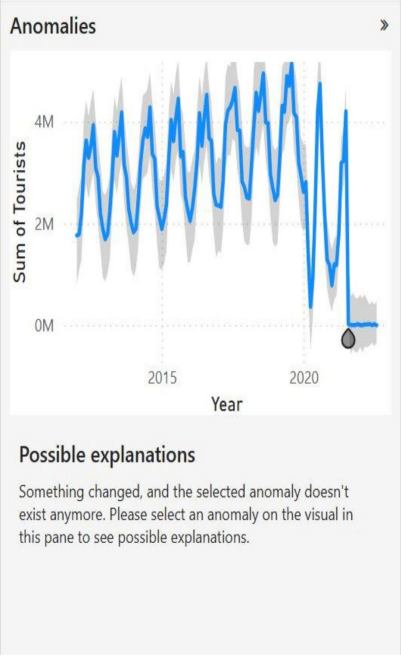
Travel is Domestic was unusually low — the strongest driver with 95% explanation strength.

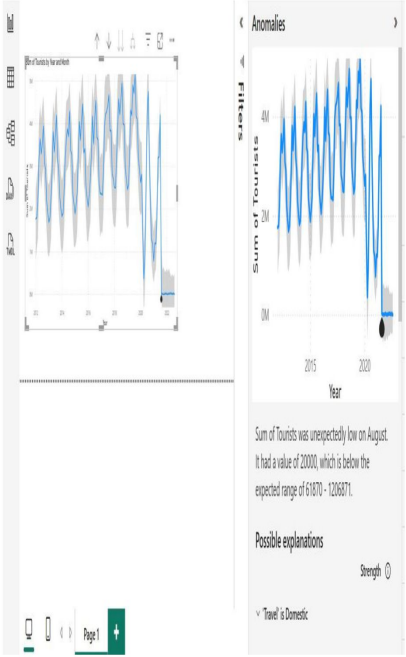
### Hotel Stays

**Strength: 31%**

Accommodation type Hotel was also unusually low, contributing 31% to the anomaly explanation.

Value of 20,000 tourists fell far below the expected range of 61,870 – 1,206,871. Domestic travel was the main explanation.



EXPLAIN BY

# Explanation Strength: Domestic (95%) & Hotel (31%)

*'Travel is Domestic' had the strongest explanation strength at 95%, followed by Hotel at 31%.*

## Travel is Domestic

**Strength: 95%**

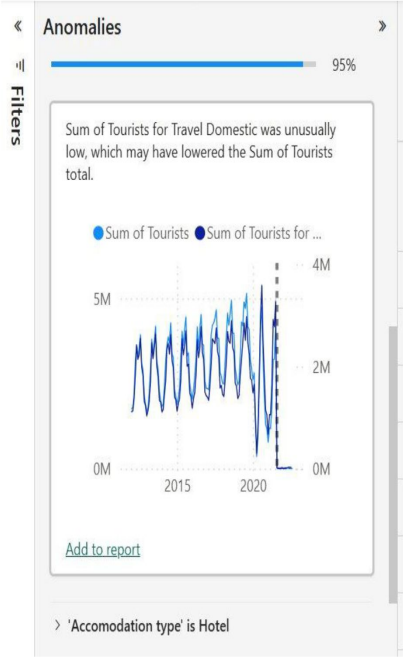
Domestic tourism was unusually low during this period — far below the seasonal expectation. This was the primary driver of the August 2022 anomaly with 95% explanation strength.

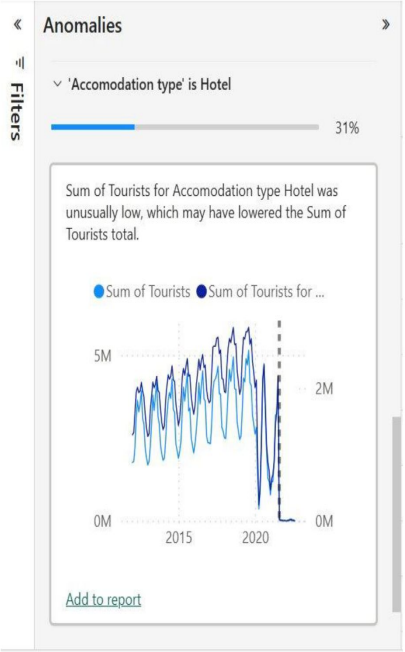
## Accommodation Type: Hotel

**Strength: 31%**

Hotel stays were also unusually low, supporting the low domestic travel finding. With 31% strength, this is a secondary but significant contributing factor to the detected anomaly.

'Travel is Domestic' had the strongest explanation strength (95%), followed by 'Accommodation type is Hotel' (31%).



# SUMMARY

*IS107 | Laboratory Work 6 — Anomaly Detection in Time-Series Data*

- ✓ Tourism data (2012–2022) showed consistent seasonal peaks mid-year and an upward trend up to 2019.
- ✓ Anomaly detection at 70% sensitivity flagged both HIGH (June 2019) and LOW (2020–2021) outliers.
- ✓ The June 2019 high anomaly was driven by unusually high Holiday Park bookings.
- ✓ The 2020–2022 low anomalies clearly reflect the COVID-19 impact on Hotel stays and International travel.
- ✓ Appending the 20-row dataset revealed a new LOW anomaly in August 2022 with Domestic travel at 95% strength.
- ✓ All AI-generated explanations were logically valid and consistent with known real-world events.